

Synthetic Data Parameter Optimization for Object Detection

Framework Contributions Summary

Theoretical
Foundations

RL-Optimized
Synthetic Data
Framework

Methodological
Innovations

Key Performance Benchmarks

mAP@50: 0.91

mAP@75: 0.78

25-40 Iterations

8-12x Efficiency

Validated Across Diverse Application Domains

Production-Ready Detectors Trained Exclusively on Synthetic Data

Ch. 2-3

MDP Formulation

GP Surrogates

BOHB Algorithm

Composite Rewards

Ch. 5-6

Multi-Objective Opt.

Validation Design